

Chapter 8 Answer Key

Introduction to Statistics — MATH& 146

Section 8.4 Exercises (Goodness-of-Fit)

8.4.1. Die rolled 120 times: $O = (18, 22, 17, 25, 19, 19)$. Test fairness at $\alpha = 0.05$.

Step 1. Hypotheses:

- H_0 : the die is fair; $p_i = 1/6$ for $i = 1, \dots, 6$.
- H_a : the die is not fair; at least one $p_i \neq 1/6$.

There are $k = 6$ categories, so $df = 5$.

Step 2. Expected counts: $E_i = 120 \cdot \frac{1}{6} = 20$ for every face. All $E_i \geq 5$. ✓

Step 3. Test statistic:

Face	1	2	3	4	5	6
O_i	18	22	17	25	19	19
E_i	20	20	20	20	20	20
$(O_i - E_i)^2/E_i$	0.20	0.20	0.45	1.25	0.05	0.05

$$\chi^2 = 0.20 + 0.20 + 0.45 + 1.25 + 0.05 + 0.05 = 2.20.$$

Step 4. Critical value at $\alpha = 0.05$ with $df = 5$ is 11.07. Since $2.20 < 11.07$, we have $p > 0.05$ and **fail to reject** H_0 .

Conclusion. There is insufficient evidence to conclude the die is unfair. The observed counts are within the range we would expect from 120 rolls of a fair die.

8.4.2. Compute the expected count for each cell in Exercise 1. What would it mean if every observed count exactly equaled its expected count?

Every expected count is $E_i = 120/6 = 20$. If every observed count exactly equaled its expected count, then $O_i - E_i = 0$ for all i , so every term in the sum is zero and $\chi^2 = 0$. This is a perfect fit to the null model: the data agree with H_0 as closely as the data possibly could. The p -value would equal 1, meaning the data give zero evidence against H_0 .

(In practice, a χ^2 of *exactly* zero is essentially never observed in real data, but the closer χ^2 is to zero, the better the fit.)

- 8.4.3.** The χ^2 statistic is always non-negative. Why? What is the minimum possible value and when does it occur?

Each term in the sum has the form $(O_i - E_i)^2/E_i$. The numerator $(O_i - E_i)^2$ is a square, so it is non-negative. The denominator E_i is a positive expected count (we require $E_i \geq 5$ for the test). The ratio of a non-negative number to a positive number is non-negative, so every term is ≥ 0 , and a sum of non-negative terms is non-negative.

The minimum possible value is $\chi^2 = 0$, achieved exactly when $O_i = E_i$ for every cell.

Section 8.5 Exercises (Independence)

- 8.5.1.** 200 students, class year (Freshman, Sophomore, Junior, Senior) $\times$ study location (Library, Dorm, Cafe). Set up hypotheses and state degrees of freedom.

- H_0 : class year and preferred study location are independent.
- H_a : class year and preferred study location are associated.

The table has $r = 4$ rows and $c = 3$ columns, so

$$\text{df} = (r - 1)(c - 1) = (4 - 1)(3 - 1) = 3 \cdot 2 = 6.$$

- 8.5.2.** For a 4×3 table with grand total $n = 500$, a cell has row total 120 and column total 75. Compute its expected count.

$$E = \frac{(\text{row total})(\text{column total})}{n} = \frac{120 \cdot 75}{500} = \frac{9000}{500} = 18.$$

- 8.5.3.** In the sleep/exercise example, identify the two cells that contribute the most to χ^2 . What does each tell you about the association?

The two largest contributions are:

- **Often / Good sleep:** contribution 12.42. Observed count 58 vs. expected 36.67. Far more people who exercise often report good sleep than independence would predict.
- **Often / Poor sleep:** contribution 10.42. Observed count 10 vs. expected 26.67. Far fewer people who exercise often report poor sleep than independence would predict.

Together, these two cells say the same thing from opposite ends: people who exercise often are pulled toward good sleep and away from poor sleep, much more strongly than independence allows. The “Often” row is the dominant source of the association.

(The cells in the “Rarely” row tell the mirror-image story: more poor sleep and less good sleep than independence predicts. The pattern is monotone across the table.)

Section 8.6 Exercises (Reading the Table)

8.6.1. For each χ^2 statistic and df , report the conclusion at $\alpha = 0.05$ and bracket the p -value.

(a) $\chi^2 = 7.1$, $df = 3$.

Row 3 critical values: 6.251 ($\alpha = 0.10$), 7.815 ($\alpha = 0.05$), 11.345 ($\alpha = 0.01$). Since $6.251 < 7.1 < 7.815$, we have $0.05 < p < 0.10$. **Conclusion at $\alpha = 0.05$: fail to reject H_0 .**

(b) $\chi^2 = 18.5$, $df = 6$.

Row 6 critical values: 12.592 ($\alpha = 0.05$), 16.812 ($\alpha = 0.01$), 22.458 ($\alpha = 0.001$). Since $16.812 < 18.5 < 22.458$, we have $0.001 < p < 0.01$. **Conclusion at $\alpha = 0.05$: reject H_0 .**

(c) $\chi^2 = 4.2$, $df = 2$.

Row 2 critical value at $\alpha = 0.10$ is 4.605. Since $4.2 < 4.605$, we have $p > 0.10$. **Conclusion at $\alpha = 0.05$: fail to reject H_0 .**

8.6.2. What happens to critical values as df increases (for fixed α)? Why does this make intuitive sense?

Critical values increase with df . Reading down any column of the table makes this clear: at $\alpha = 0.05$, the critical value rises from 3.841 at $df = 1$ to 18.307 at $df = 10$.

The reason: χ^2 is a sum of squared standardized discrepancies, one per cell. More cells means more terms in the sum, so the statistic naturally takes larger values *even when H_0 is true*, just from random sampling noise. To keep the false-rejection rate fixed at α , the cutoff has to move further right as we add more terms.

A useful mental model: each cell can contribute a small random bump to χ^2 . With one cell, a single bump of size 4 is unusual. With ten cells, a total of 4 across all of them is easy to get by chance. So the same numerical χ^2 value is more or less impressive depending on how many cells are summed.

Chapter 8 Review

1. Botanist claims germination ratio 4:3:2:1 for varieties A, B, C, D. Plants 200 seeds; observes $A = 88$, $B = 60$, $C = 35$, $D = 17$.

(a) Hypotheses.

- H_0 : the four varieties germinate in the ratio 4:3:2:1; that is, $p_A = 0.4$, $p_B = 0.3$, $p_C = 0.2$, $p_D = 0.1$.
- H_a : the germination proportions differ from this claim.

(b) Expected counts. The ratio sums to $4 + 3 + 2 + 1 = 10$, so the claimed probabilities are 0.4, 0.3, 0.2, 0.1. Multiplying by $n = 200$:

$$E_A = 80, \quad E_B = 60, \quad E_C = 40, \quad E_D = 20.$$

All $E_i \geq 5$. ✓

(c) Test statistic.

Variety	A	B	C	D
O_i	88	60	35	17
E_i	80	60	40	20
$(O_i - E_i)^2/E_i$	$\frac{64}{80} = 0.800$	0	$\frac{25}{40} = 0.625$	$\frac{9}{20} = 0.450$

$$\chi^2 = 0.800 + 0 + 0.625 + 0.450 = 1.875.$$

(d) Decision. Critical value at $\alpha = 0.05$ with $df = 3$ is 7.815. Since $1.875 < 7.815$, we have $p > 0.05$ and **fail to reject** H_0 .

Conclusion. There is insufficient evidence to reject the botanist's claimed 4:3:2:1 germination ratio. The data are consistent with that claim.

2. Employees: gender $\times$ work arrangement, $n = 400$.

(a) Hypotheses and df.

- H_0 : gender and preferred work arrangement are independent.
- H_a : gender and preferred work arrangement are associated.

The table has $r = 2$ rows and $c = 3$ columns, so $df = (2 - 1)(3 - 1) = 2$.

(b) Expected counts. Use $E_{rc} = (\text{row total})(\text{column total})/n$. Row totals are 180, 220; column totals are 170, 130, 100.

	Remote	Hybrid	Office
Male	$\frac{180 \cdot 170}{400} = 76.5$	$\frac{180 \cdot 130}{400} = 58.5$	$\frac{180 \cdot 100}{400} = 45.0$
Female	$\frac{220 \cdot 170}{400} = 93.5$	$\frac{220 \cdot 130}{400} = 71.5$	$\frac{220 \cdot 100}{400} = 55.0$

Row sums: $76.5 + 58.5 + 45.0 = 180$ ✓ and $93.5 + 71.5 + 55.0 = 220$ ✓. All $E_{rc} \geq 5$. ✓

(c) Test statistic.

$(O - E)^2 / E$	Remote	Hybrid	Office
Male	$\frac{(80-76.5)^2}{76.5} = 0.160$	$\frac{(60-58.5)^2}{58.5} = 0.038$	$\frac{(40-45)^2}{45} = 0.556$
Female	$\frac{(90-93.5)^2}{93.5} = 0.131$	$\frac{(70-71.5)^2}{71.5} = 0.031$	$\frac{(60-55)^2}{55} = 0.455$

$$\chi^2 = 0.160 + 0.038 + 0.556 + 0.131 + 0.031 + 0.455 = 1.371.$$

(d) Decision. Critical value at $\alpha = 0.05$ with $df = 2$ is 5.991. Since $1.371 < 5.991$, we have $p > 0.05$ and **fail to reject** H_0 .

Conclusion. There is insufficient evidence of an association between gender and preferred work arrangement. The two largest contributions come from the Office column, where men chose office work slightly less than expected and women slightly more, but the gap is well within sampling noise.

3. Student computes $\chi^2 = 3.2$ with $df = 1$.

(a) Row 1 critical values: 2.706 ($\alpha = 0.10$), 3.841 ($\alpha = 0.05$), 6.635 ($\alpha = 0.01$).

At $\alpha = 0.05$: $3.2 < 3.841$, so **fail to reject**. Not significant.

At $\alpha = 0.10$: $3.2 > 2.706$, so **reject**. Significant at the 10% level.

This illustrates that “significance” depends on the chosen α ; the same statistic can flip the decision when the threshold moves.

(b) The minimum χ^2 needed to reject at $\alpha = 0.01$ with $df = 1$ is $\chi_{0.01,1}^2 = 6.635$.

4. Ice cream sales and drowning deaths: $\chi^2 = 18.4$, $df = 1$, $p < 0.001$. Journalist writes “Ice cream causes drowning.”

The journalist has confused association with causation. The chi-squared test of independence detects a statistical relationship between two variables; it does not identify which (if either) causes the other, nor does it rule out a third variable explaining both.

The most plausible confounder here is **warm weather** (or equivalently, summer / hot temperatures). In hot weather, ice cream consumption rises because people want a cold treat, and swimming activity rises because people want to cool off, which in turn increases the number of drownings. Both ice cream sales and drowning deaths are driven by the underlying variable of temperature; neither causes the other.

This is a textbook example of **spurious correlation through a lurking variable**. A causal claim would require an experimental design (which is obviously not ethical here) or a careful observational analysis that controls for temperature, time of year, and water-exposure rates.

5. Big picture: compare χ^2 goodness-of-fit to the one-sample z -test for a proportion.

Both tests address the same general question: does an observed categorical outcome match a claimed model? They differ in the number of categories.

One-sample z -test for a proportion ($k = 2$ categories): the variable has exactly two outcomes (success / failure). The claim is a single number p_0 , and the test statistic is $z =$

$(\hat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$. Either tail of the standard normal can give evidence against H_0 , depending on the direction of the alternative.

Goodness-of-fit ($k \geq 2$ categories): the variable has any number of categories. The claim is a whole probability vector $(p_1, p_2, \dots, p_k)$, and the test statistic $\chi^2 = \sum (O - E)^2/E$ combines all categories into a single number. Only the right tail of the χ^2 distribution gives evidence against H_0 , because the statistic measures total mismatch.

When to use each:

- Use the z -test when there is exactly one binary outcome of interest and the alternative is directional (e.g., “is p greater than 0.5?”). It gives a signed test statistic, so it can answer one-sided questions naturally.
- Use goodness-of-fit when there are three or more categories, or when the question is about the *overall shape* of a distribution rather than one parameter. With $k = 2$ categories and a two-sided alternative, the goodness-of-fit test is mathematically equivalent to the squared z -test ($\chi^2 = z^2$), and gives the same conclusion.

What goodness-of-fit offers that the z -test cannot: the ability to test a distribution with many categories at once. The z -test can only ask “is one proportion equal to a claim?”; the goodness-of-fit test can ask “does this entire distribution match the claim?” For something like M&M color proportions or accident counts by day of the week, the z -test simply cannot handle the question, while χ^2 handles it in one calculation.

The goodness-of-fit test is the natural generalization of the proportion z -test from two categories to k categories. The cost of that generalization is loss of direction; we no longer know *which* category is high or low until we look at the individual cell contributions.